DEPARTMENT OF WORKS & HIGHWAYS

MATERIAL SCIENCE & TECHNOLOGY BRANCH

Materials Testing Laboratory

√ Port Moresby ☐ Mt. Hagen, ☐ Lae, ☐ Annex Lab.

Date:

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DRY DENSITY / MOISTURE CONTENT RELATIONSHIP - AS1289.5.1.1 & 5.2.1

Client : ---------- Asset No.: ----------

Project: ---------- Job No.: ----------

Location: ---------- Mat'l Source / Pit No.: ----------

Material Description: ---------- Date : ----------

STANDARD / MODIFIED MOULD VOL. cm^3 MOULD No. MOULD TYPE _

Sample Condition (NMC): ....(%)

Soil Type (LL) :.....(%) Others: .....

Curing Time: .....(Hrs) Initial Time: .....Date: ...../...../..... Final Time: .....Date...../...../....

Materials retained on 19mm (AS.sieve + 19mm materials replaced by 19mm - 5mm materials) YES / NO ☐

Number of layers							
Number of blows / layer							
Water added	ml						
Approximate increase in moisture content	%						
M1 Mass of compacted soil and mould	g						
M2 Mass of mould	g						
M3 Mass of compacted soil (M1 - M2)	g						
M4 Volume of mould	cm 3						
M5 Wet density $\frac{(M3)}{(M4)}$	t/m3						
M6 Dry density $\frac{(M5) \times 100}{100 + (M12)}$	t/m3						
Container number							
M7 Mass of wet soil and container	g						
M8 Mass of dry soil and container	g						
Mc Checked mass - M8	g						
M9 Mass of container	g						
M10 Mass of water (M7) - (M8)	g						
M11 Mass of dry soil (M8) - (M9)	g						
M12 Moisture content. $\frac{(M10) \times 100}{(M11)}$	%						

Note: Check mass (Mc) must be used on all moisture content calculations

Tested by: _ Checked by: _ _ _ _ _ _ _ _ _ _ _ _ _ _

Date: _ Date: _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _

Authorised by: AS(Material Science & Technology Branch)

FORM No.S-5.1.1

Revised Worksheet: 08/04/2022 lbh